INCH-POUND
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MILITARY PERFORMANCE SPECIFICATION

BLAST CLEANING MACHINES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 2 types of dry blast cleaning machines for surface cleaning, surface conditioning and peening of ferrous and nonferrous metals using a blasting media.

1.2 Classification. The types of machines covered by this specification are of the following types and sizes:

Type I	Type II
Cabinet, Stationary	External hand-held gun (see 3.3.2.1)
Size A (see 3.3.1.2)	
Size B (see 3.3.1.2)	
Size C (see 3.3.1.2)	

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: AFLCMC/WNZEB, Robins AFB GA 31098. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks of the exact revision listed below form a part of this specification to the extent specified herein.

COMMERCIAL ITEM DESCRIPTIONS

A-A-59316	Abrasive Materials; for Blasting

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-808	Finish, Protective and Codes for Finishing Schemes for Ground and Ground Support Equipment

(Copies of these documents are available online at https://quicksearch.dla.mil/ or https://assist.dla.mil).

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE AMS2431/6	Peening Media, Glass Shot
SAE AMS-STD-595	Colors Used in Government Procurement
SAE J444	Cast Shot And Grit Size Specifications For Peening And Cleaning

(Application for copies should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale PA 15096). Electronic copies may be obtained from https://www.sae.org/servlets/index.

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

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3.2 Design, Materials, and manufacturing processes. Unless otherwise specified, the design, materials, and manufacturing process selection shall be the prerogative of the contractor as long as all articles submitted to the government fully meet operating, interface, ownership and support, and operating environment requirements specified.

3.3 Operating requirements. The machine shall be designed for simplicity of operation and safety to operating personnel and shall permit easy accessibility of components for normal maintenance and servicing. A suction or pressure type feed shall supply blasting media to the blast gun where high pressure air under controlled conditions propels the blasting media at high velocity in a uniform and consistent stream to the parts being cleaned or conditioned. The machine shall be capable of blasting continuously for periods in excess of 7 minutes. The blast shut-off time required for automatic recharging of the abrasive hopper shall not exceed 30 seconds. The machine shall be designed to accommodate blasting media of glass shot (SAE AMS/2431/6) Nominal G110 to G230 Steel Shot and Steel Grit conforming to SAE-J444 and aluminum oxide, silicon carbide, garnet and emery conforming to A-A-59316.

3.3.1 Type I.

3.3.1.1 Operational requirements. The Type I blast machine shall be so designed so that the operator directs the blast media upon the work piece while standing outside the cabinet and extending his arms into the cabinet through protective openings. This Type I machine design shall include an exhaust and dust collector system for maintaining a dust free climate within the cabinet. A reclaim system shall remove reusable blasting media from the dust stream and return the blasting media to the abrasive hopper. The dust shall be collected in a metal enclosed dust collector to include a Hepa type filter system. The Type I machine shall include a cabinet, exhaust and dust collector system, reclaim system, abrasive hopper, blast gun and hose assembly, air and media operating controls, air pressure regulator and gage, moisture separator and an instructional panel. The unit shall be capable of regulating a pneumatic supply pressure of 125 pounds per square inch gage to an operating pressure of 20 to 100 pounds per square inch gage. When blasting at 80 pounds per square inch gage with size 13 glass beads, the machine shall produce a cleaned path not less than 1 inch wide at a rate of 1 lineal foot in 10 seconds. When blasting at 80 pounds per square inch gage with 26 grit aluminum oxide, the machine shall produce a cleaned path not less than 1 inch wide at a rate of not less than 4 lineal feet in 10 seconds. The noise of the operators shall not exceed 84 decibel A slow scale.

3.3.1.2 Design requirements. The cabinets shall be provided in three optional sizes and be constructed from sheet metal:

Size A - Work area 4 feet wide by 3 feet deep by 3 feet high
Size B - Work area 5 feet wide by 4 feet deep by 3 feet high
Size C - Work area 6 feet wide by 6 feet deep by 5 feet high

The size of the B and C cabinet door shall not be less than 52 inches. It shall be of rigid construction and prevent escape of dust or blast media. The work floor shall be capable of supporting a 500-pound load with a maximum allowable deflection of ¼ inch. The floor shall be supported within the cabinet in a manner that it can be easily removed and replaced. An observation window shall be located above the arm ports and at a height as to permit the operator full view of the area covered by the abrasive discharge. The glass shall be perpendicular to the operator's line of sight. The cabinet shall be illuminated by external fluorescent or incandescent light. The light shall provide an allowable light intensity of 200-foot candles at the center of the

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cabinet at the work level and 150-foot candles at the corners of the cabinet at the work level. A loading car track and extension assembly shall support a minimum workload of 150 pounds per square foot of car loading area. An exhaust and filter system shall maintain the inside of the cabinet free of dust while the operator is blast cleaning. The unit shall be provided with a reclaim system, abrasive hopper of not less than 40 pounds of size 3 glass beads with provisions for draining of the beads. The unit shall be provided with a blast gun with hose assembly, an air pressure regulator and gage, a moisture separator and instructional panel.

3.3.2 Type II.

3.3.2.1 Operational requirements. The Type II blast machine shall be designed for cleaning objects that are not suitable for cleaning a cabinet type machine. The blast gun shall be so designed that the operator can clean flat, round, cylindrical and square objects without loss of blasting media to the atmosphere. The Type II machine design shall include a recovery system to return spent blasting media from the workpiece to the unit and a reclaim system to separate reusable blasting media from the air stream and return it to the abrasive hopper. The useable blasting media and dust shall be collected in a Hepa type filter system. The blast generating unit shall be mounted on casters to afford easy maneuverability. The unit shall be capable of regulating a pneumatic supply pressure of 125 pounds per square inch gage to an operating pressure of 20 to 100 pounds per square inch gage. It shall be capable of discharging a constant, evenly distributed stream of media on a material surface having a minimum curvature of 12-inch radius and reclaim, clean and recycle the blasting media during the operation. When blasting at 80 pounds per square inch gage with size 13 glass beads, the machine shall produce a cleaned path not less than 1 inch wide at a rate of 1 lineal foot in 10 seconds. When blasting at 80 pounds per square inch gage with 26 grit aluminum oxide, the unit shall produce a cleaned path not less than 1 inch wide at a rate of 4 lineal feet in 10 seconds.

3.3.2.2 Design requirements. The Type II unit shall be equipped with a cyclone type reclaim system that separates reusable abrasive from the spend abrasive returned from the workpiece by the recovery system. The reclaimed abrasive shall be returned to the abrasive hopper. The hopper shall have a capacity of not less than 5 pounds of size 3 glass beads, have provisions for draining the beads and all components shall be accessible for repair. The unit shall be provided with a blast gun and 20 feet of hose assembly. The abrasive blast cleaning nozzles shall be equipped with an operating valve which must be held open manually. The blast machine shall be mounted on industrial type casters to afford maneuverability by one man. The unit shall be provided with an air pressure regulator with gage, moisture separator, instructional panel, brushes and storage compartment.

3.3.3 Power output. The unit shall be operable from a 110/220 volt, single phase, 60 Hz power outlet.

3.3.4 Finish and painting. The machine shall be finished and painted in accordance with MIL-HDBK-808, F-100. The Type I cabinet interior shall be white, color number 37886 of SAE AMS-STD-595, and the exterior shall be gray, color number 16473 of SAE AMS-STD-595. The Type II machine shall be gray color number 16473 of SAE AMS-STD-595.

3.4 Ownership and support requirements. The unit shall meet the following logistics requirements.

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3.4.1 Reliability. Achieve at least 6000 hours mean time between failure (MTBF).

3.4.2 Safety. Protect personnel from the blast operations with no detected emissions from the blast operations.

3.4.3 Maintainability. Provide easy access to components requiring routine or frequent maintenance or adjustment. The units shall require no additional lubrication or preventive maintenance during its normal service life.

3.5 Interface requirements.

3.5.1 Ownership and support requirements verification. The unit shall meet the following logistics requirements.

3.5.2 Reliability. The unit shall be capable of 6,000 hours between failures (MTBF). Reliability calculations shall be based on the total amount of hours operated without failure divided by the number of times the unit is unable to perform as required in this specification.

3.5.3 Safety. The unit shall be inspected for hazardous burrs, nicks, sharp edges, foreign materials, or other imperfections that pose physical danger to an installer/operator. Inspect the unit for exposed, frayed, unsecured, or otherwise improperly protected circuits that pose electrical danger to an installer/operator.

3.5.4 Maintainability. Upon completion of reliability evaluation, verify that the unit does not require preventative maintenance such as lubrication, cleaning or adjustment. Demonstrate or prove other evidence that key design and assembly characteristics contribute to the ease of maintenance and prevention of common mishandling, reassembly, or reconnection errors.

3.6 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. The pre-production tests shall consist of all the tests described under 4.4 except the individual acceptance tests specified in 4.4.12.

4.3 Conformance inspection. Each production item shall be subjected to an acceptance test consisting of the following tests as described under 4.4.

a. Examination of the product (see 4.4.1)
b. Mechanical inspection (see 4.4.2)

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c. Functional check (see 4.4.3)
d. Pressure test (see 4.4.4)
e. Dielectric strength test (see 4.4.5)
f. Individual acceptance test (see 4.4.12)
g. Noise test (Type I) (see 4.4.13)

4.4 Test methods.

4.4.1 Examination of the product. The blast machine shall be examined to determine compliance with the requirements of this document. The examination shall be conducted using a manufacturing developed check list that lists each specification characteristic, method of verification and results obtained.

4.4.2 Mechanical inspection. A mechanical inspection of all components and parts of the machine for defects and damage shall be conducted. All pertinent data concerning condition, defects in workmanship, and damage shall be recorded.

4.4.3 Functional check. All mechanical parts of the blast machine shall be checked to determine if each operates freely and functions properly. The functional check shall be conducted using a manufacturer developed list that shall show functions checked, how checked and results obtained.

4.4.4 Pressure test. Each component and circuit of the blast machine that operates under positive pressure in normal service shall be hydrostatically (or pneumatically or with carbon dioxide gas at the contractor's option) pressure tested at 1 ½ times working pressure for not less than 10 minutes. Distortion, damage, or leaks resulting from the pressure testing shall be cause for rejection of the components affected.

4.4.5 Dielectric strength test. The electrical system shall be subjected to a dielectric strength test at a potential equal to twice the working voltage plug 1,000 volts for a period of one minute. Any arcing, flash-over or charring of the insulation shall cause rejection of the machine.

4.4.6 Performance test. The blast machine shall be subjected to a 4-hour performance test using blasting media conforming to the requirements of 3.3. During the test period, the machine shall be used to perform cleaning tasks that will demonstrate the cleaning capability of the unit. The contractor will provide the exhibits to be cleaned, which shall be irregular shapes and shall present cleaning problems typical to the application of this type of equipment. While blasting at 80 pounds per square inch gage on a flat surface, the machine shall produce a cleaned path not less than one inch wide at the rate of one lineal foot in 10 seconds using size 13 glass beads and at the rate of 4 lineal feet in 10 seconds using 26 grit aluminum oxide. In addition to the cleaning tasks performed, Almen strips shall be peened to show compliance with Table I as specified.

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TABLE I. Performance Characteristics.

Bead Size	Air pressure at Regulator (psig)	Time (in seconds)	Peen. Intensity (min),)+.0005 A)
3	40	22	0.0060
3	60	22	0.0100
3	80	22	0.0140
8	40	22	0.0025
8	600	22	0.0040
8	80	22	0.0045
12	40	22	0.0015
12	60	22	0.0020
12	80	22	0.0025

4.4.7 Reclaim system test. The reclaim system test shall be performed using Size 8 glass beads as the blasting media. The blasting media shall be sieved to ensure a sieve analysis conforming to the limits set forth in SAEJ444. A minimum of 5 pounds of the blasting media shall be cycled through the blast machine at least 10 times while blasting at 80 pounds per square inch gage against a flat steel plate. The media will then be drained from the machine and weighed; the weight being recorded. After weighing, the media will again be sieved and the media retained on the sieve shall weight not less than 80 percent of the weight recorded above. Values less than 80 percent shall cause rejection of the machine.

4.4.8 Exhaust and dust collector test (Type I). During the 4-hour performance test of the Type I blast machine, the exhaust and dust collector system shall be observed for proper operation. Certification shall be considered as evidence that the exhaust system replaced the air within the cabinet at least 10 times per minute. However, a dust free and leak proof system shall still be required and proven to exist.

4.4.9 Recovery and dust collector test (Type II). During the 4-hour performance test of the Type II blast machine, the recovery and dust collector system shall be observed for proper operation. The contractor shall demonstrate the recovery system to be 90 percent efficient. The blast gun brushes shall not allow escape of blast media to the atmosphere. A dust free climate shall exist around the blast machine. The mechanical filter shaker shall be operated, and the dust-tray removed and emptied.

4.4.10 Work floor test (Type I). The work floor shall be loaded at the center with 500 pounds and the maximum deflection measure. The deflection shall not exceed ¼ inch.

4.4.11 Lighting system test (Type I). Using a General Electric Type 213 light meter or equivalent, the light intensity shall be checked at the work floor with the access door closed. The light meter shall be placed on the center of the work floor and moved to each corner in turn. The minimum reading shall be 150-foot candles in each corner and 200-foot candles at the center.

4.4.12 Individual acceptance test. Each production item blast machine shall be subjected to a one-hour performance test using blasting media of size 13 glass beads for 30 minutes and 26 grit aluminum oxide for 30 minutes. During the test period, the blast machine shall perform cleaning tasks typical to the application of this type of equipment.

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4.4.13 Noise test (Type I). The noise level at the operator's position shall be measured using a general sound level meter. A noise level in excess of 84 db A slow scale shall cause rejection.

5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or ordering data (see 6.2i). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory).

6.1 Intended use. The machines covered by this document are intended for use in cleaning, surface conditioning, and peening of Aerospace Ground Equipment (AGE) and aircraft components.

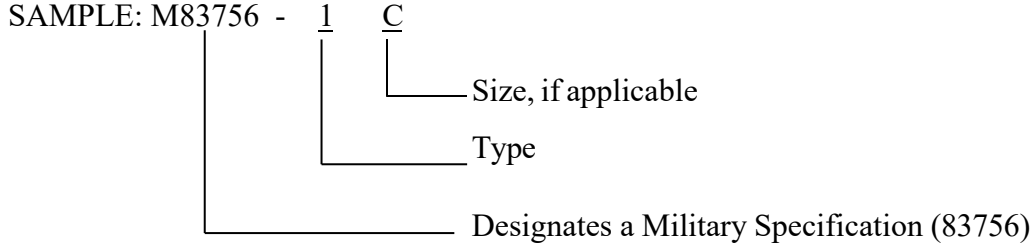
6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. When first article is required (see 3.1).
c. Type and size as specified (see 1.2).
d. Packaging requirements (see 5.1).

6.3 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include: contractor past performance, Government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those processes, labor skill and training, and the uniformity of measuring processes and techniques.

6.4 Recovered materials. Materials collected and recovered from solid waste and reprocessed to become a source of raw materials, as compared to virgin raw materials.

6.5 Part identifying number (PIN). The following PIN is developed for Government purposes and use with this CID. It does not constitute a requirement for the contractor:



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6.6 Key Words.

Cabinet
Exhaust
Filter
Gage
Glass shot
Peening
Pressure
Reclaim system
Suction
Work floor

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
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Preparing Activity:
Air Force - 184

Reviewers:
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Air Force - 110

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